Analytical flexibility in large language model based reviews

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Pre-registration

Preamble

Given that some of the text in the preregistration will be used in the final manuscript, we opt to write the description below in the past tense for the sake of efficiency. Readers should also note that we may allude to specific figures in the text (e.g., “Figure XX”). Note that this text represents placeholder text which will be completed in the final study manuscript

Introduction

A growing issue within research is the increasing volume of scholarly publications and the corresponding demand for peer review (Adam, 2025). This can create *reviewer fatigue*, a state in which scholars increasingly decline review invitations as they become overextended by the sheer magnitude of requests (Breuning et al., 2015).

As another observation, the adoption of large language models (LLMs) has accelerated significantly in recent years. Chatterji et al. (2025) reported that by July 2025, ChatGPT alone reached approximately 700 million users, representing roughly 10% of the global population. Consequently, LLMs have become a pervasive subject across various scientific disciplines, influencing the entire research lifecycle (Schmidgall et al., 2025). Psychological research is no exception; the field has been profoundly affected by LLM applications. Some studies apply LLMs as a tool, such as to model cognition (Binz et al., 2025; Milička et al., 2024) Another stream applies LLMs as autonomous agents , for example to generate data under certain constraints (Dillion et al., 2023; Hewitt et al., 2024).

Among the many suggestions of how LLMs may be used by researchers, one suggestion has been as a tool to alleviate the issue of reviewer fatigue. LLMs could for example assist in predicting paper quality and acceptance rates or recommend reviewers to ensure a better alignment of expertise (Pendyala et al., 2025). Moreover, LLMs can author reviews completely automated. Some research has reported promising results regarding fully LLM-authored reviews, suggesting they can facilitate expeditious and constructive feedback (Idahl & Ahmadi, 2025). Furthermore, evidence indicates that LLM-authored reviews can achieve relative parity with human experts in terms of depth (Liang et al., 2023).

The promise of LLM-authored reviews is not absolute. Researchers have identified shortcomings when LLMs function as autonomous review agents, such that they might exhibit self-bias, preferring text which was written by the judges themselves (Y. Liu et al., 2024) – while struggling to reliably detect errors (Son et al., 2024), generating false-positive claims (R. Liu & Shah, 2023) or falling short of adversarial prompt attacks (Collu et al., 2025).

While the source of this divergence in literature remains unclear, it may be contingent upon specific, yet often undisclosed, methodological choices. Scholars have begun to recognize that varying configurations significantly influences outcomes. This includes, though is not limited to, the selection of model versions (Stamatogiannakis et al., 2025), tuning of hyperparameters such as the temperature (Li et al., 2025), and implementation of specific prompt techniques (Sclar et al., 2024) .If these novel degrees of freedoms are ignored, we risk repeating the shortcomings of conventional research and end up with low replicability once again. Consequently, it remains unclear to what extent current findings in LLM-authored research have already been influenced by these undisclosed variables.

Recent research has found analytic flexibility in silicon sampling (Cummins, 2025), such that data features like correlations are subject to sensitivity under different parameter configurations, and within annotation tasks (Baumann et al., 2025; McLaren et al., 2026). These findings warrant an investigation into the influence of analytic flexibility on the domain of LLM-authored reviews. Consequently, in the current study we adopted an analogous framework to examine whether a comparable degree of flexibility exists in the context of LLM-authored reviews. We systematically manipulated several configurational settings, including *model architecture*, *hyperparameter* (temperature or reasoning effort), the implementation of *chain of thought* (CoT) prompting (Kojima et al., 2023; Wei et al., 2023) and the intermediate step of *note-taking* (Robertson & Koyejo, 2025). These dimensions were selected because they represent unavoidable researcher degrees of freedom; decisions regarding model architecture, the hyperparameters, the user prompt and a potential intermediate step, as for example note-taking, are fundamental to any LLM-based workflow.

The primary goal of this study was to determine the extent to which different choices affect the quality of LLM-authored reviews. To assess the quality of an LLM-authored review we applied the LLM-as-a-judge paradigm (Chen et al., 2024). In particular we applied the *score-only* fine-tuned Llama-3.1-8B model provided by Sadallah et al. (2025), which judged each single comment of the review's weakness section on the four dimensions *actionability,* grounding & *specificity, verifiability*, and *helpfulness*.

The secondary goal of this study was to determine the extent to which different choices affect the generation of LLM-authored reviews, independently of external quality assessments. For this we monitored the operational success of an LLM-authored review.

As detailed below, this study was guided by two primary research questions. All of them examined the extent to which analytic flexibility within LLM configurations influences research outcomes. Specifically, our primary research question investigated whether LLM-authored reviews exhibit significant sensitivity in their quality across different configurations. The second research question examined whether the rate of successfully generated reviews shows sensitivity across different configurations.

Method

Below, we briefly describe our data generation steps. However, we also invite the reader to inspect our fully preregistered code for precise specifications on all aspects of data generation. The implementations of these represent the precise and formal preregistration rather than our loose description here.

Sample

The source materials for the generation workflow of the LLM-authored reviews consisted of the original manuscripts publicly available from the publicly available ICLR 2023 peer-review dataset hosted on OpenReview. The exact code to obtain these original manuscripts can be inspected on our GitHub repository[1]. While our initial target was a sample of 100 papers (see "Hardcoded definitions", line 4), two

[1] https://github.com/hiiamlars/analytical_flexibility_llm_reviews/blob/main/data_generation/review_data_crawl.ipynb

papers could not be successfully parsed from the raw PDF into text format using our planned parsing method (GROBID) – a limitation also reported by Xu et al (2024). As a secondary measure, text extraction for these two papers was attempted using the DPT-2 model from LandingAI (REF), which likewise proved unsuccessful (for both these steps see the segment “Parsing”). Consequently, these papers were excluded, resulting in a final test sample of 98 papers. We maintained the inclusion criterion established in previous studies about LLM-authored reviews, for example Xu et al. (2024), that each paper must have at least three human-written reviews (see the segment “ICLR Data”). Although this condition was not strictly necessary for the current study, it was retained to ensure the dataset remains compatible with this and other analyses conducted on the ICLR dataset.

Review generation

In phase two of the workflow, we used the original manuscripts to generate the LLM-authored reviews. Again, our exact code for this step can be inspected on the GitHub repository[2].

We deployed various LLMs from OpenAI, Groq and Google across a range of parameters; each distinct combination of model and parameters was instructed to generate a review of each manuscript. The baseline user prompt was taken from Robertson and Koyejo (2025) and instructed the model to draft an ICLR-style review, including the strengths and weaknesses of the underlying manuscript (see the segment “Setup definitions”, lines 25-82 and 93-124, for details of the models, parameters and the prompt and see the segment “Configuration grid” for how all possible combinations are constructed). Because the subsequent pipeline required the clear and structured isolation of weaknesses within the generated text, we integrated a Pydantic response schema directly into the generation process. This schema enforced that each generated review would, different to the instruction given via the baseline user prompt, provide the strengths and weaknesses of the underlying manuscript in distinct, structured lists (see the “Setup definitions” segment, lines 133-141). To test subsequent phases of the workflow we included a simulation

[2] https://github.com/hiiamlars/analytical_flexibility_llm_reviews/blob/main/data_generation/llm_review_generation.ipynb

mode which would produce an artificial review adhering to the response schema (see the "Setup definitions", lines 6-9, and "Generation client" segment).

Utilizing the enforced response schema, we extracted the weakness section of each LLM-generated review and segmented it into single comments.[3] Concurrently, we assessed whether each LLM-authored review was generated successfully (see the segment "Success metric").

Review judgement

In a third phase, we applied the score-only fine-tuned judge developed by Sadallah et al. (2025) to each individual comment[4]. This allowed us to obtain quality scores across the four dimensions: actionability, grounding & specificity, verifiability, and helpfulness. Once again, we utilized a Pydantic response schema to enforce structural reliability from the evaluation model. For this step, however, the prompt instructions and the Pydantic response schema were entirely aligned (see the segment "Setup definitions", lines 31-210). Again, we included a simulation mode to test subsequent phases of the workflow (see the segments "Setup definitions", lines 6-9, and "Score client"). Lastly, we parsed the raw response of the model into their final quality scores.[5]

Research Questions

We addressed two primary research questions in this study.

RQ 1

Do LLM-authored reviews exhibit systematic variation in their review quality as a function of configurational settings, namely the variation of the model architecture, the hyperparameter (temperature or reasoning effort), the chain of thought instruction (present vs. absent) and the note-taking strategy (present with a faithful strategy vs. absent), as measured by the average helpfulness per configurational group?

[3] https://github.com/hiiamlars/analytical_flexibility_llm_reviews/blob/main/data_generation/llm_review_segmentation.ipynb
[4] https://github.com/hiiamlars/analytical_flexibility_llm_reviews/blob/main/data_generation/llm_review_judgments.ipynb
[5] https://github.com/hiiamlars/analytical_flexibility_llm_reviews/blob/main/data_generation/llm_review_post_processing.ipynb

RQ 2

Does the generation of LLM-authored reviews exhibit systematic variation in its success as a function of configurational settings, namely the variation of the model architecture, the hyperparameter (temperature or reasoning effort), the chain of thought instruction (present vs. absent) and the note-taking strategy (present with a faithful strategy vs. absent), as measured by the operational success rate per configuration group?

Analytic Plan

Below, we briefly describe our analyses steps. However, we also invite the reader to inspect our fully preregistered code for precise specifications on all aspects of the analyses. The implementations of these in R represent the precise and formal preregistration rather than our loose description here.

Prior knowledge of the dataset

To uphold the principle of the holdout sample and prevent researcher bias (Baldwin et al., 2022) the final analysis script (analyses.Rmd)[6] was developed and validated using the simulated data before being executed on the actual LLM-authored reviews and subsequent judgements. This procedure ensured the final sample and the statistical results remained unknown to us while the analyses pipeline was prepared.

Scoring algorithm

Review quality

The score-only fine-tuned LLM-judge evaluated each comment on the following four dimensions (Sadallah et al., 2025, p. 28994):

- **Actionability**: Actionability assesses the extent to which a comment offers practical guidance for the authors. It evaluates how concrete and detailed the suggestions are, and how easily they can be translated into specific actions.

[6] https://github.com/hiiamlars/analytical_flexibility_llm_reviews/blob/main/analyses/analyses.qmd

- **Grounding & Specificity**: This aspect measures the extent to which a review comment is anchored to specific parts of the paper. It evaluates whether the comment clearly identifies the relevant text region and explicitly states what requires improvement.
- **Verifiability**: This aspect evaluates whether a review comment contains a claim (i.e., a subjective opinion) and how well it is substantiated. The first step is to determine whether the comment includes any claims. If it does, we assess the extent to which the reviewer justifies or supports the claim through logical reasoning, common knowledge, or references.
- **Helpfulness**: Helpfulness evaluates the overall usefulness of a comment to the authors, aggregating the previous three aspects. It is a subjective measure that reflects the overall value of the review comment to the authors. Therefore, this aspect integrates our three preceding, primary aspects and is a subjective rating by the annotators.

To obtain an indicator of quality for each LLM-authored review, scores across comments of the same LLM-authored review were averaged (see "Transformation and calculation", line 174). We inspected the helpfulness score specifically as the primary outcome for review quality as this dimension is seen as an aggregate one; the other three dimensions are only considered as supplementary.

Operational success

During the review generation process, each combination of paper and configuration either had resulted in some generated review which also adhered to our predefined schema, or it failed on any of both criteria. Each LLM-authored review had already been assigned a success status within the data generation phase, and thus this score did not needed to be averaged at this stage (see "Transformation and calculation", line 177).

Confidence intervals

After we had prepared the helpfulness score and the operational success rate both for each LLM-authored review, we bootstrapped both metrics within each configuration group. Some others calculated their bootstrap with 500 iterations (Simonsohn et al., 2020). However as the our study design follows closer Cummins (2025), we used the same amount of 2,000 iterations to generate 95% confidence intervals (α = .05) for each of the measures. After bootstrapping we were able to summarize the data into scores and confidence intervals per configuration group (see "Bootstrap", lines 98-99 and 186-242).

Analyses

RQ 1 Sensitivity of Review Quality

The estimated quality of LLM-authored reviews ranged from XX to XX. Whether LLM-authored reviews exhibit sensitivity to their quality was assessed with the main quality metric (the average helpfulness of generated reviews per configuration group). Within the specification curve analysis, non-overlapping confidence intervals across configurations would signify a significant influence of configuration parameters on the review's quality. The systematic overlap of confidence intervals across the specification curve would suggest a lack of significant configuration influence on the review's quality. The specification curve plot for the quality metric can be found in Figure XX.

RQ 2 Sensitivity of Operational Success Rate

The estimated operational success rate ranged from XX to XX. Whether the generation of LLM-authored reviews exhibit sensitivity to configurational choices was assessed with the operational success rate (the rate of successful review generations per configuration group). Within the specification curve analysis, non-overlapping confidence intervals across configurations would signify a significant influence of configuration parameters on the success of the generation process. Conversely, a systematic overlap of confidence intervals across the specification curve would suggest a lack of significant configuration influence on the operational success rate. The specification curve plot for the operational success metric can be found in Figure XX .

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